

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED**CO4:**

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions; while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion

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Compare star topology and Bus topology in terms of cost, Reliability and performance.

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Cost: star topology is generally more expensive because every device is connected individually to a central device such as a switch or Hub. This requires more network cables and networking hardware. In contrast, bus topology uses a single backbone cable to which all devices are connected. Therefore, it requires less cabling and fewer devices, making it cheaper and easier to install.

Reliability: star topology provides better reliability. If one computer or one connecting cable fails, the remaining devices can continue communication, because each device has an independent connection to the central switch or hub.

Performance: star topology generally provides better performance because communication is managed through the central networking device. This helps

reduce collisions and provides better control over data transmission. It is therefore more suitable when several devices need to communicate regularly.

2. Compare Mesh Topology and Star Topology in terms of fault tolerance.

Mesh topology provides very high fault tolerance because each node has connections to multiple other nodes. Consequently, there can be several possible paths between devices. If one connection fails, data can be transmitted through another available path.

Star topology provides moderate fault tolerance. Each individual device has a separate connection to a central switch or hub. Therefore, if one computer or its cable fails, the other computers are normally unaffected.

However, the central device is a major dependency. If the central switch or hub fails, the communication between the connected devices can be disrupted. Thus, although star topology is easier to manage, its fault tolerance is lower than that of a mesh topology.

2° Compare Bus Topology and Mesh Topology in terms of installation complexity.

In a bus topology, all devices share a single backbone cable. Only minimal connections are required, so installation requires less cabling and fewer ports. This makes bus topology suitable for small networks

where simplicity and low installation cost are important.

Mesh topology has a completely different structure. Each node is connected to multiple other nodes, and in a full mesh arrangement, every node may have a direct connection to every other node. As the number of devices increases, the number of required connections increases rapidly.

- More cables are required.
- More ports and networking interfaces are needed.
- Installation takes more time.
- Configuration becomes more complicated.
- Troubleshooting becomes more difficult.

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Compare Hybrid Topology and star topology in terms of scalability.

In a hybrid topology, different sections of an organization can use different topologies according to their requirements. For example, one department may use a star topology while another section may use a different topology. These sections can then be connected to form a larger hybrid network.

Star topology is also scalable, but its expansion is dependent on the capacity of its central switch or hub. Every device needs to connect to this central switch device. If all available ports are already occupied, additional hardware is required to connect more devices.

So, star topology provides good scalability within the limits of its central device, whereas hybrid topology provides greater flexibility and scalability because different topologies can be combined and new network segments can be added independently.

5. Compare Star, Mesh, Bus and Hybrid Topologies in terms of Maintenance.

Star topology: It is easiest to maintain because each device has a separate connection, making faults easy to identify and isolate.

Mesh topology: is difficult to maintain because it has many connections. Troubleshooting these connections can take more time.

Bus topology: is harder to maintain because a fault in the main backbone can affect the entire network and may be difficult to locate.

Hybrid topology: has moderate maintenance complexity. It is flexible, but managing different combined topologies requires careful monitoring and skilled administration.